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INTELLIGENT NETWORK PERFORMANCE OPTIMISATION AND SECURITY AUTOMATION WITH VPN INTEGRATION

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MOTIVATION, PROBLEM STATEMENT & OBJECTIVES

Existing Network Challenges

- Cybersecurity threats
- VPN congestion & latency
- Poor QoS under heavy traffic
- Static network management

Why Intelligent Optimisation?

- Enterprise & IoT growth
- Secure VPN communication
- Adaptive traffic control
- Real-time threat detection

Primary Objectives

- Network optimisation
- Security automation
- Intelligent VPN management
- QoS improvement

A hybrid AI-based framework integrating Fuzzy Logic, Artificial Neural Networks (ANN), and Genetic Algorithms (GA) for intelligent network optimisation and security automation.

LITERATURE REVIEW

AI-based optimisation frameworks for IoMT networks inspired the use of ANN prediction and GA-based optimisation for intelligent network management [1].

Intelligent network operations using AI and Large Language Models were surveyed in [2], while genetic algorithm-based optimisation techniques were explored in [3].

Zero-Trust Security using Software-Defined Perimeter (SDP) over VPN was introduced for secure remote access in [4], and VPN traffic classification techniques were analysed in [5].

Genetic Algorithm approaches for VPN routing, reliability, and optimisation were discussed in [6] and [11].

Security assessment using Bayesian Networks and Fuzzy Inference Systems was presented in [7], while ANN and GA-based threat and malware detection methods were studied in [8–9].

QoS management using fuzzy logic and optimisation techniques such as PSO and hybrid GA scheduling methods were proposed in [10–15].

PROPOSED HYBRID FRAMEWORK

Key Features

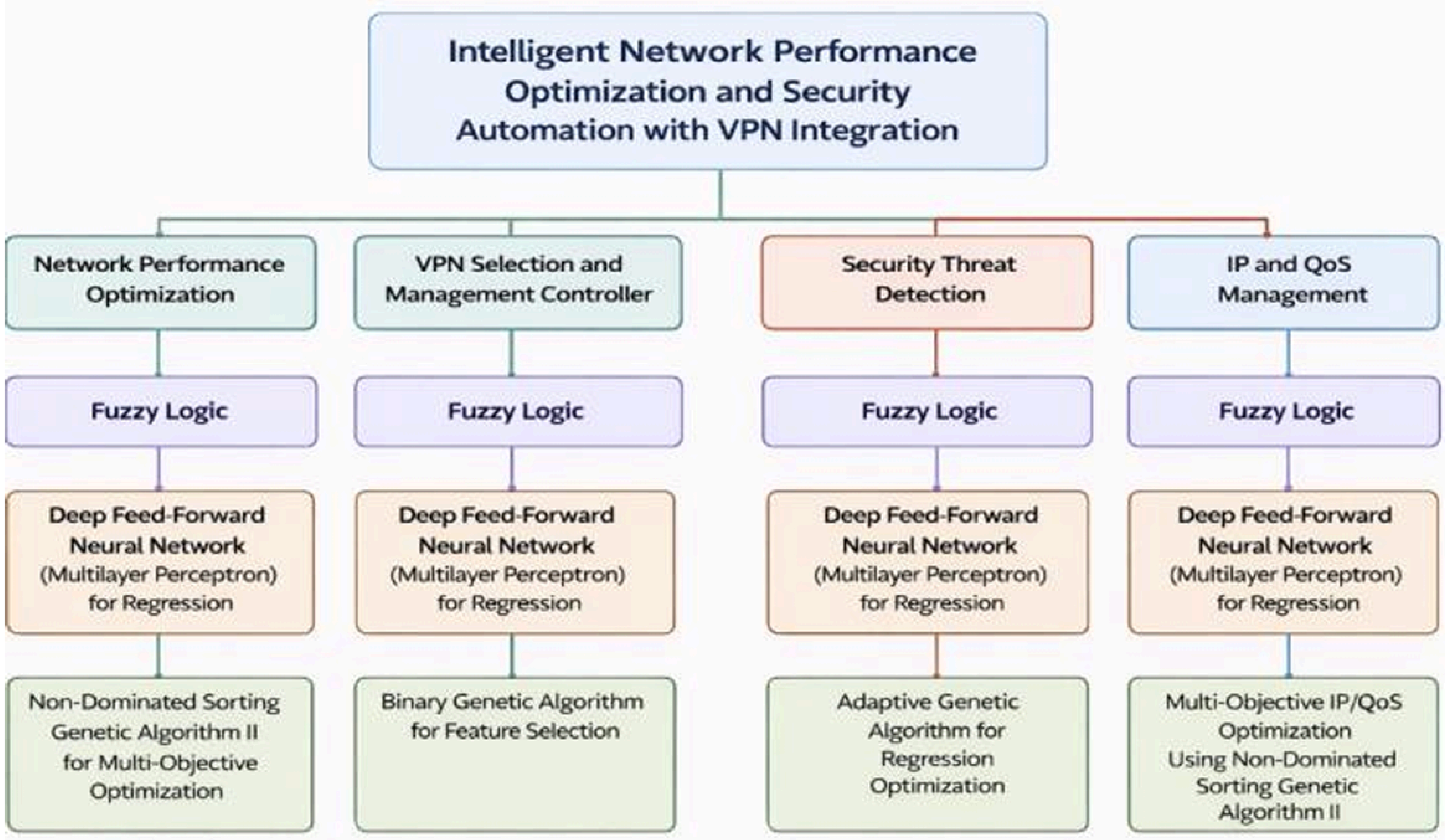
✔️ Real-time intelligent optimisation

✔️ Adaptive decision-making

✔️ Automated security management

✔️ QoS-aware traffic control

✔️ Dynamic VPN optimisation



INPUT AND OUTPUT PARAMETERS USED

1. NETWORK PERFORMANCE OPTIMISATION

Inputs
- Network Latency
- Bandwidth Utilisation
- Packet Loss Rate
- CPU Usage
- Memory Usage

Outputs
- Network Performance Index
- Optimisation Priority

2. VPN SELECTION AND MANAGEMENT

Inputs
- Server Load
- Geographic Distance
- Server Reliability
- Encryption Strength
- Connection Cost

Outputs
- VPN Selection Priority
- Connection Quality Score

3. SECURITY THREAT ASSESSMENT

Inputs
- Failed Login Attempts
- Traffic Anomaly Score
- Suspicious Port Activity
- Data Transfer Anomalies
- Geographic Risk Factors

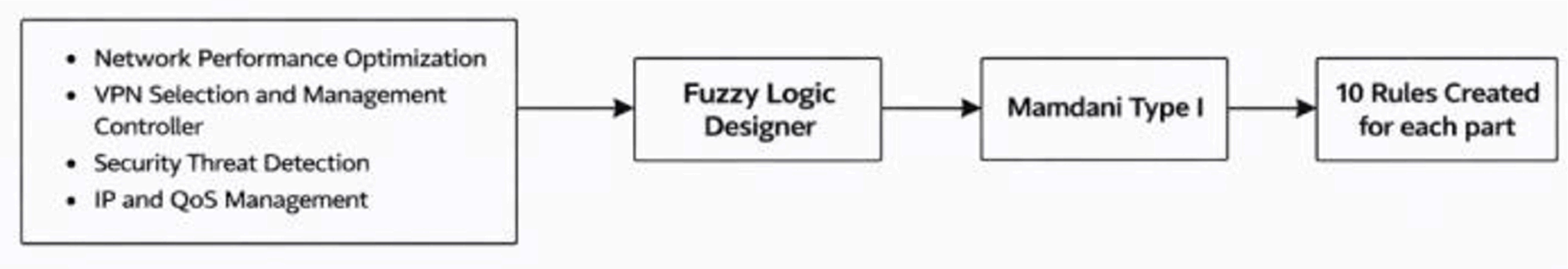
Outputs
- Threat Level
- Response Priority

4. IP AND QOS MANAGEMENT

Inputs
- IP Traffic Load
- Packet Loss Rate
- Latency
- Jitter
- IP Reputation Score

Outputs
- QoS Level
- Traffic Control Action

FUZZY LOGIC-BASED DECISION SYSTEM



Horizontal Workflow

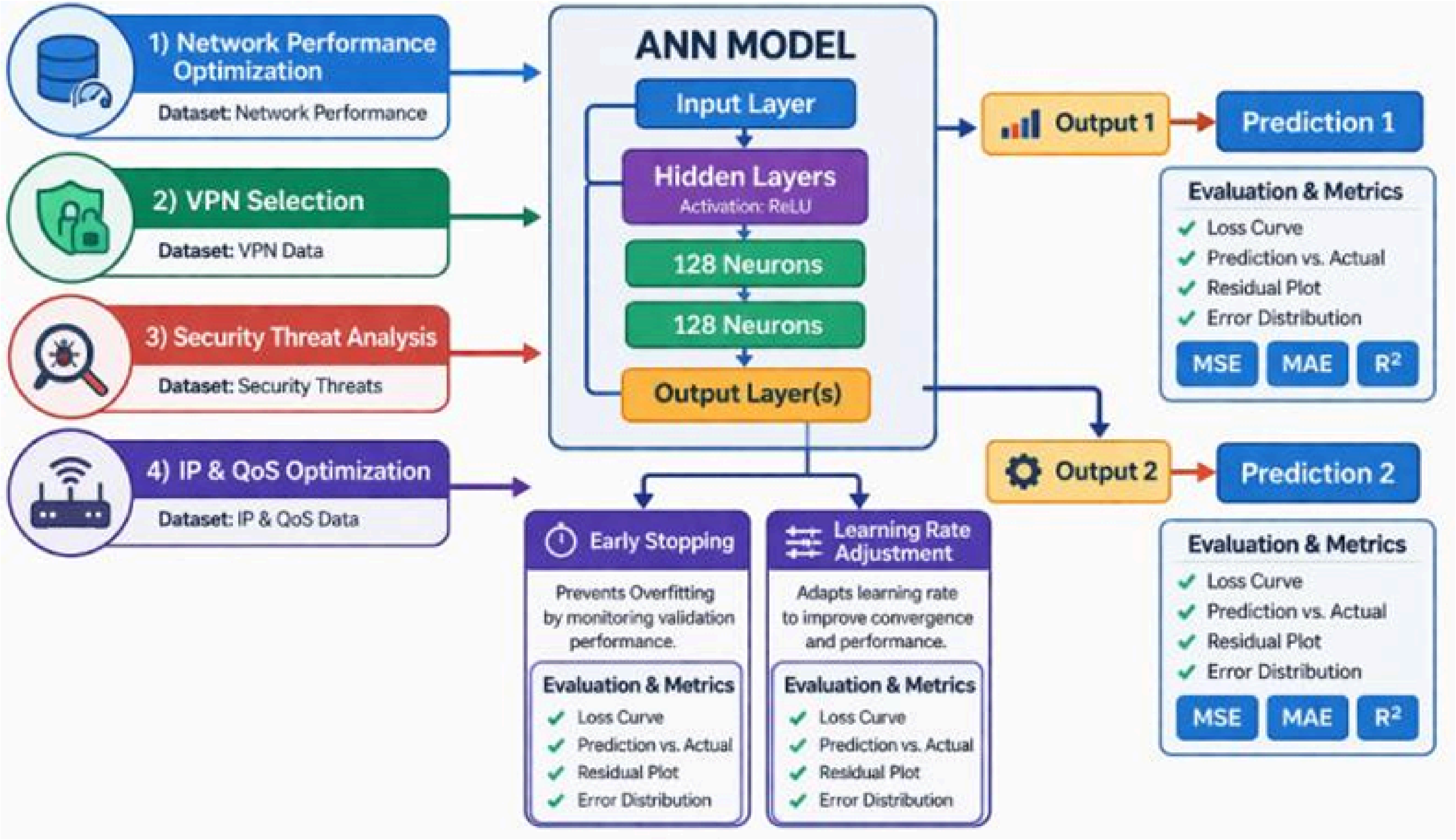
Input Parameters

- Membership Functions
- Fuzzy Rule Evaluation
- Inference Engine
- Optimised Network Decisions

Fuzzy Controllers

- Network Performance Optimisation
- VPN Selection & Management
- Security Threat Assessment
- IP & QoS Management

ARTIFICIAL NEURAL NETWORK (ANN)



ANN Features

- Deep Feed Forward ANN
- ReLU Activation Function
- Adam Optimizer
- MSE Loss Function
- Early Stopping Technique

Network Data → ANN Prediction → Intelligent Decisions

ANN EQUATIONS

- Neuron Activation (Linear Combination)

$$z_j = \sum_{i=1}^{n} w_{ij}\, x_i + b_j$$

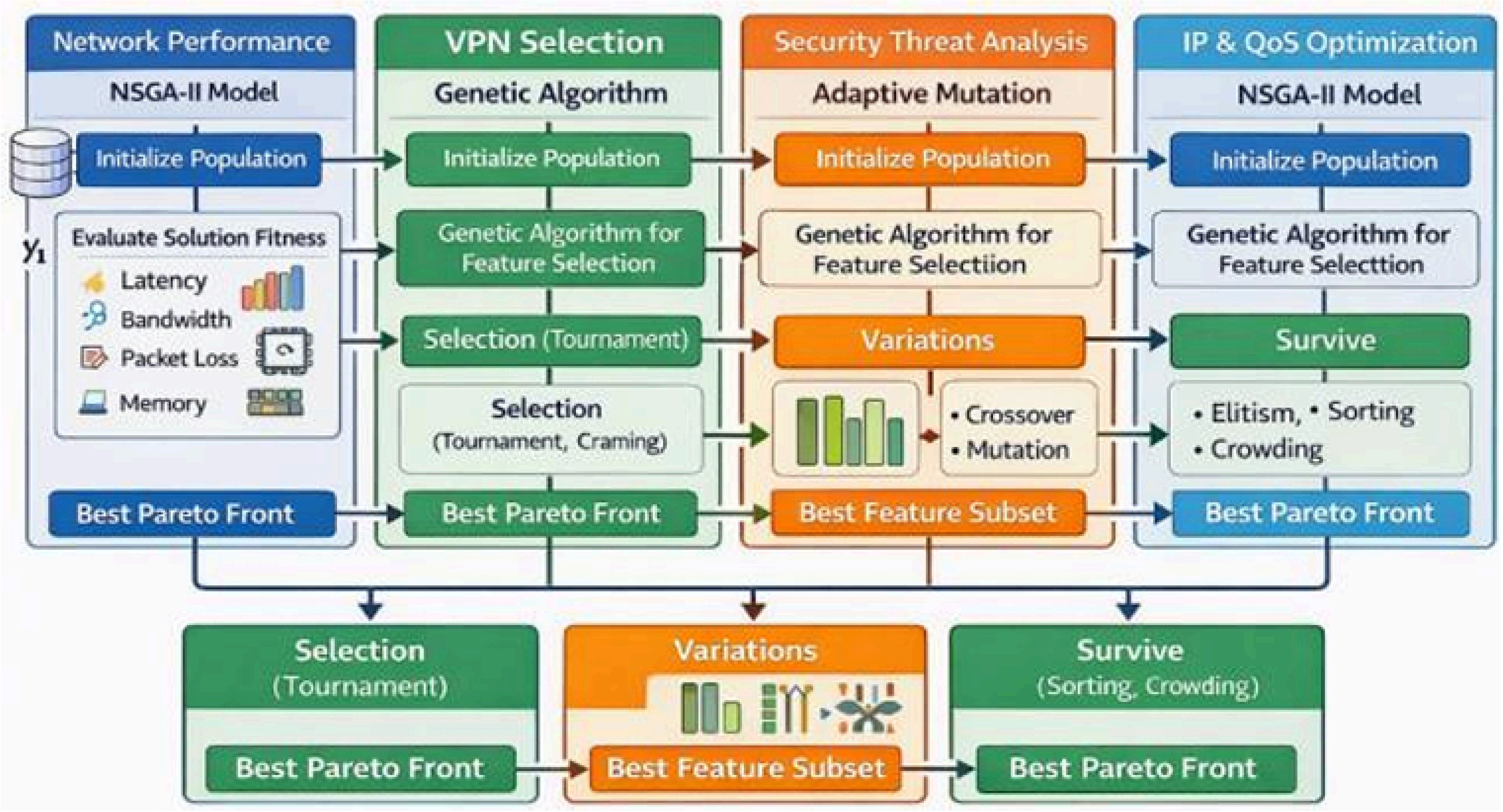
- ReLU Activation Function

$$f(zj) = \max(0, zj)$$

- Mean Squared Error (MSE) Loss

$$\mathcal{L}_{MSE} = \frac{1}{N} \sum_{k=1}^{N} (\, y_k - \hat{y}_k)^2$$

GENETIC ALGORITHM OPTIMISATION



GA WORKFLOW

Initial Population

- Fitness Evaluation
- Selection
- Crossover
- Mutation
- Optimised Solution

GENETIC ALGORITHM EQUATIONS

1. Network Performance Index (NPI)

$$\mathrm{NPI}(x) = 100 \sum_{i=1}^{5} w_i \, s_i(x)$$

2. Priority Cost Function

$$P(x) = \sum_{i=1}^{5} \alpha_i \, p_i(x)$$

3. GA Fitness Function

$$\mathrm{fit}(g) = \begin{cases} \dfrac{1}{|\mathcal{S}(g)|} \displaystyle\sum_{j \in \mathcal{S}(g)} |\rho(x_j, y)|, & |\mathcal{S}(g)| > 0, \\ 0, & \text{otherwise,} \end{cases}$$

4. Threat Prediction and Error Minimisation

$$\hat{y}^{(k)} = x^{(k)\top}\theta, \mathrm{MSE}(\theta) = \frac{1}{N} \sum_{k=1}^{N} (y^{(k)} - \hat{y}^{(k)})^2$$

5. QoS Optimisation Function

$$\mathrm{QoS}(z) = 100 \sum_{i=1}^{5} \beta_i \, q_i(z)$$

6. Cost Function

$$C(z) = \gamma_1 \frac{PL}{10} + \gamma_2 \frac{\text{Latency}}{200} + \gamma_3 \frac{\text{Jitter}}{50} + \gamma_4 \max\left(0, \frac{1000 - \text{Throughput}}{1000}\right) + \gamma_5 \max\left(0, \frac{BW - 80}{20}\right)$$

RESULTS: FUZZY LOGIC RESULTS

Subsystem	Key Parameters & Rules	Output
Network Performance	Latency, BW, CPU, Mem R1–2 High, R3–7 Med, R8–10 Low	High/Medium Performance
VPN Selection	Load, Distance, Reliability Active in R8–10 (Low load)	Secure VPN Activation
Threat Assessment	Traffic, Loss, Jitter Gradual risk levels	Normal → Critical
IP & QoS	Load, Loss, Latency Active in stable condition	QoS Enabled

- Improved Network Performance
- Secure VPN & Threat Mitigation
- Stable QoS-Aware Traffic Control
- Improved Threat Detection & Stability

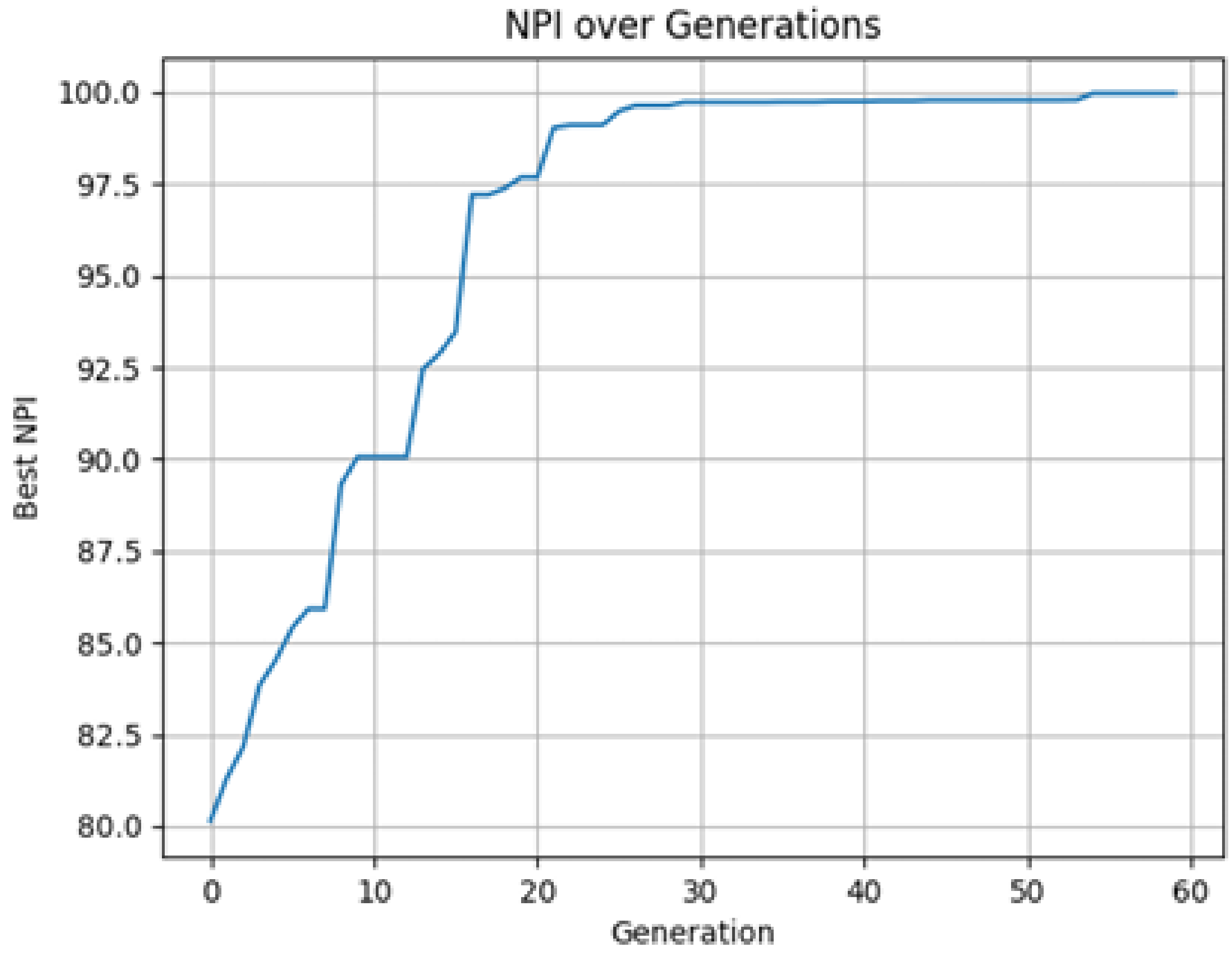
RESULTS: ANN

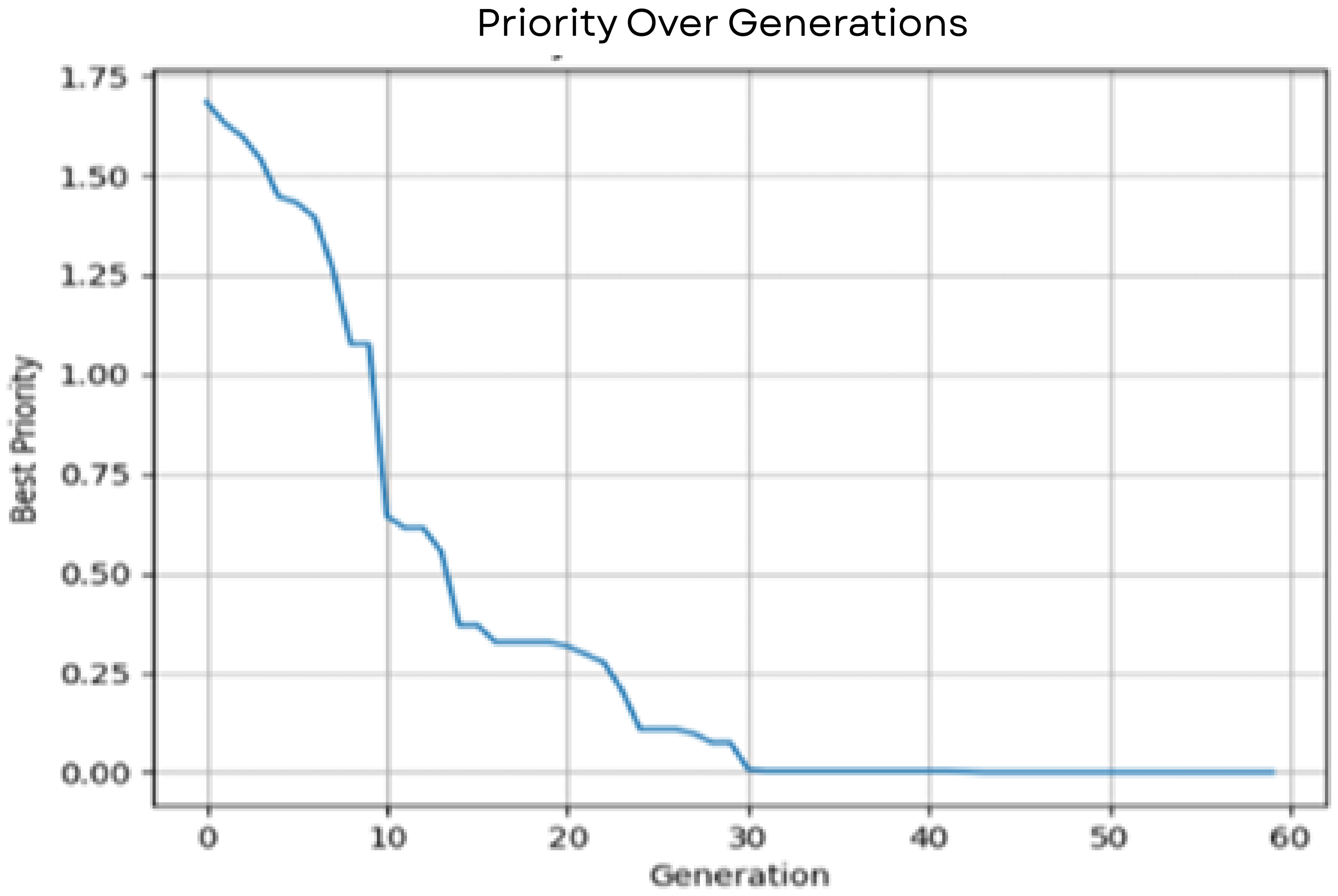
Model / Output	R^2 Score	MAE	MSE	RMSE	Accuracy (%)
Network Performance Index	0.9978	0.515	0.4387	0.6623	99.78
Optimisation Priority	0.8738	0.428	0.2855	0.5343	87.38
VPN Selection Priority	0.9981	0.5902	0.5634	0.7506	99.81
Connection Quality Score	0.9689	0.2453	0.0989	0.3144	96.89
Threat Level	0.9912	0.0772	0.0093	0.0963	99.12
Response Priority	0.925	2.397	8.8852	2.9808	92.5
QoS Level	0.998	0.3827	0.2621	0.512	99.8
Traffic Control Action	0.9964	0.0592	0.0054	0.0736	99.64

- High Prediction Accuracy
- Reduced Error & Faster Convergence
- Improved QoS & Threat Prediction

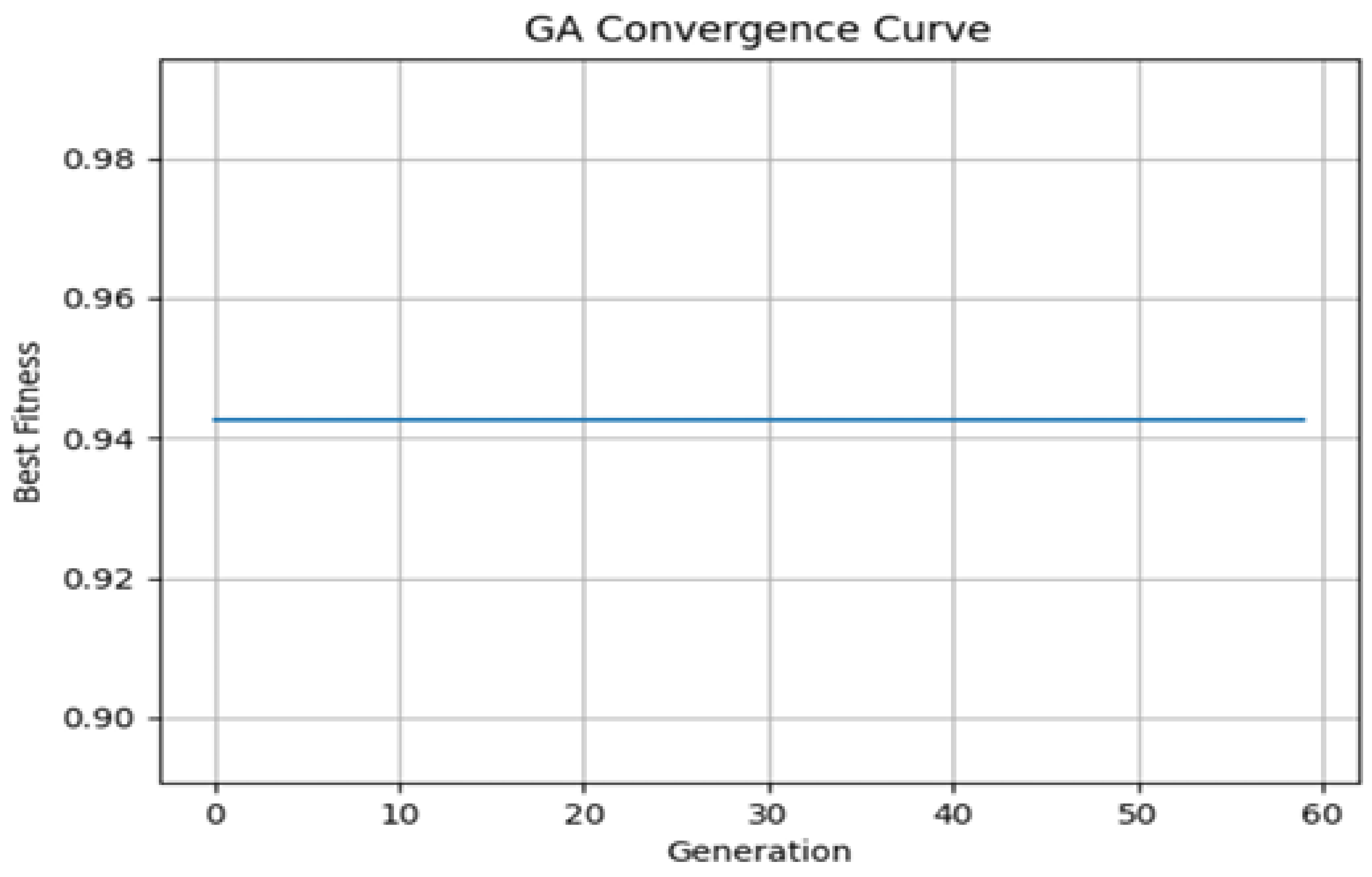
RESULTS: GENETIC ALGORITHM RESULTS

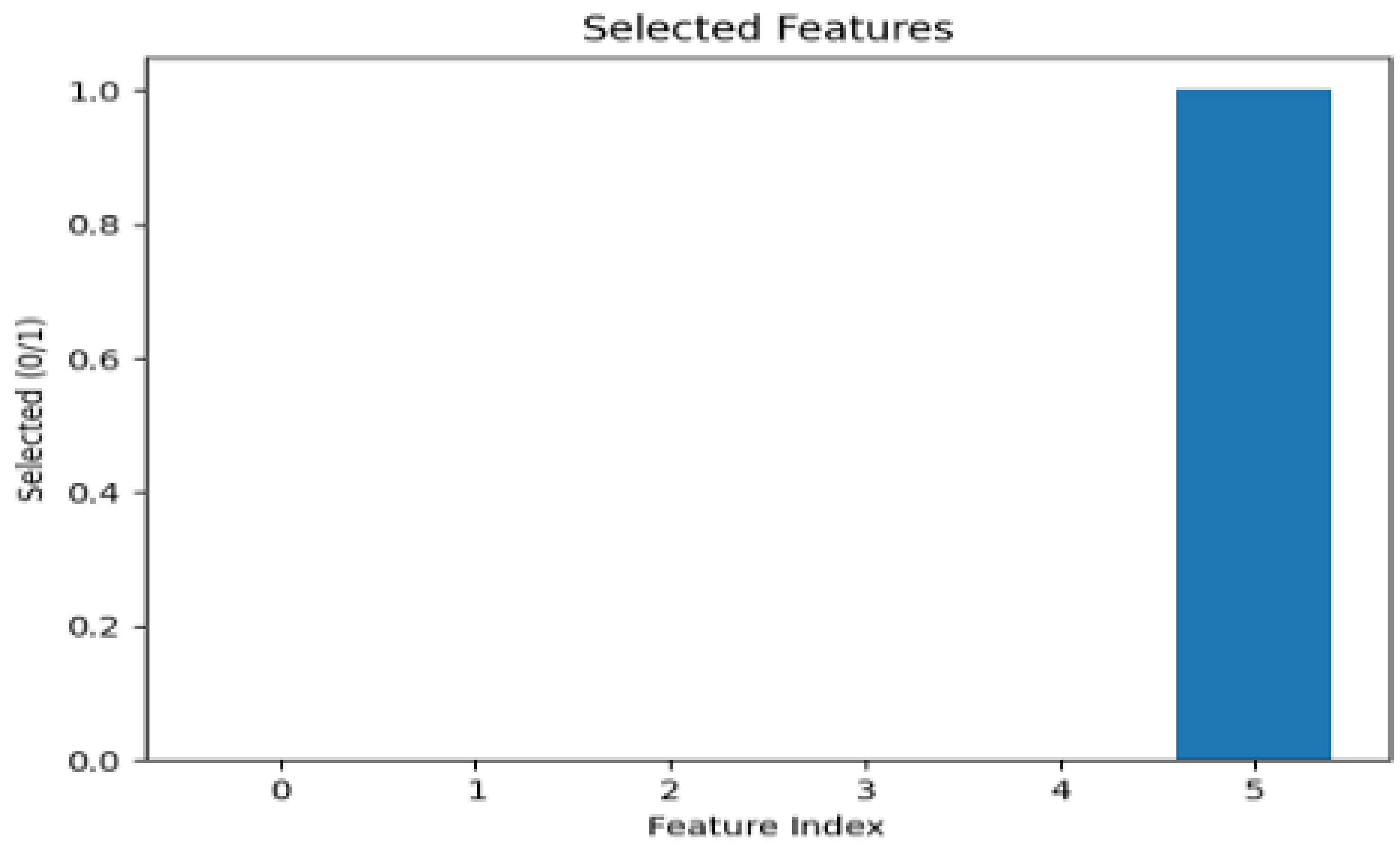
NETWORK PERFORMANCE OPTIMISATION



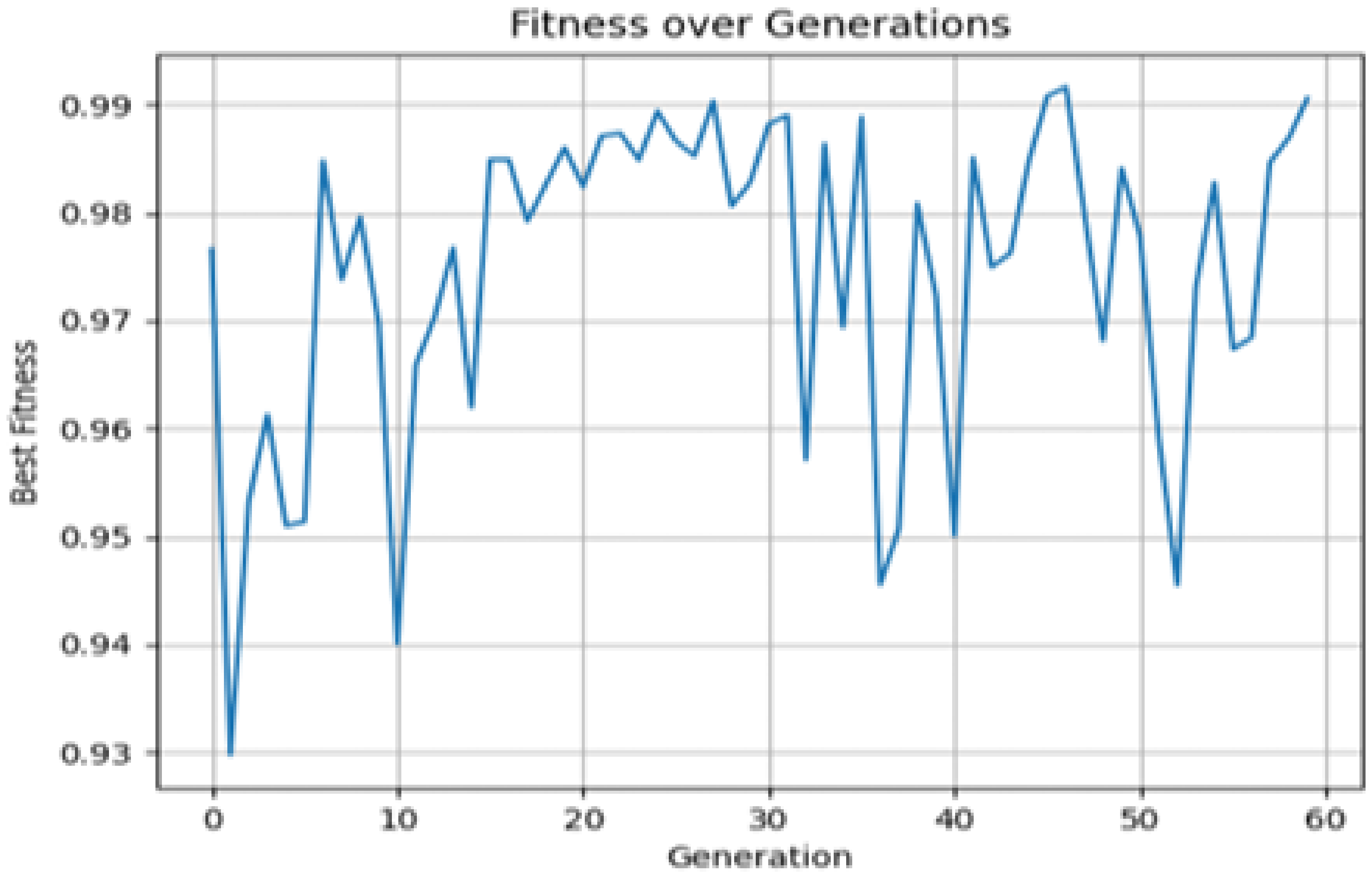


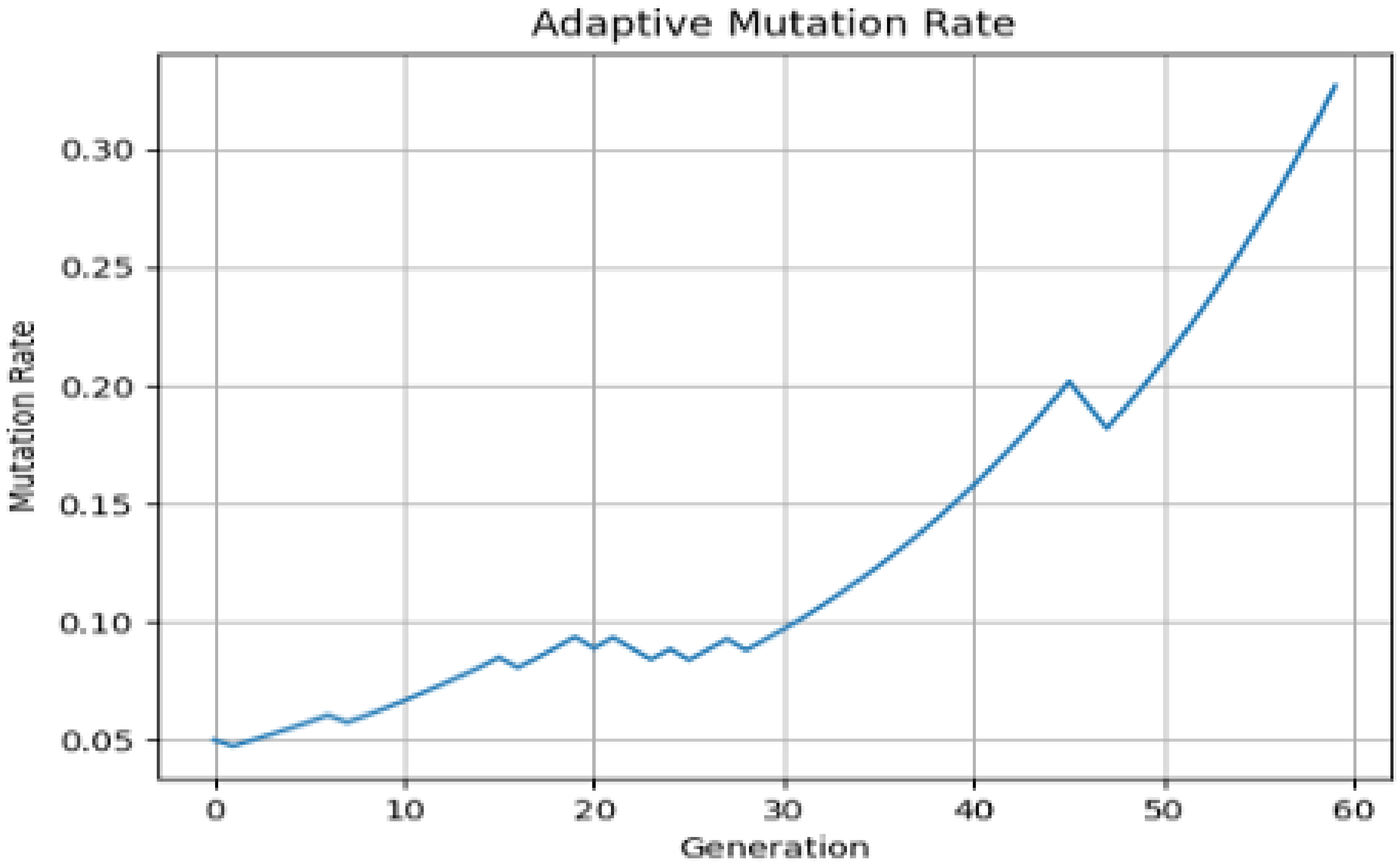
VPN SELECTION AND MANAGEMENT CONTROLLER



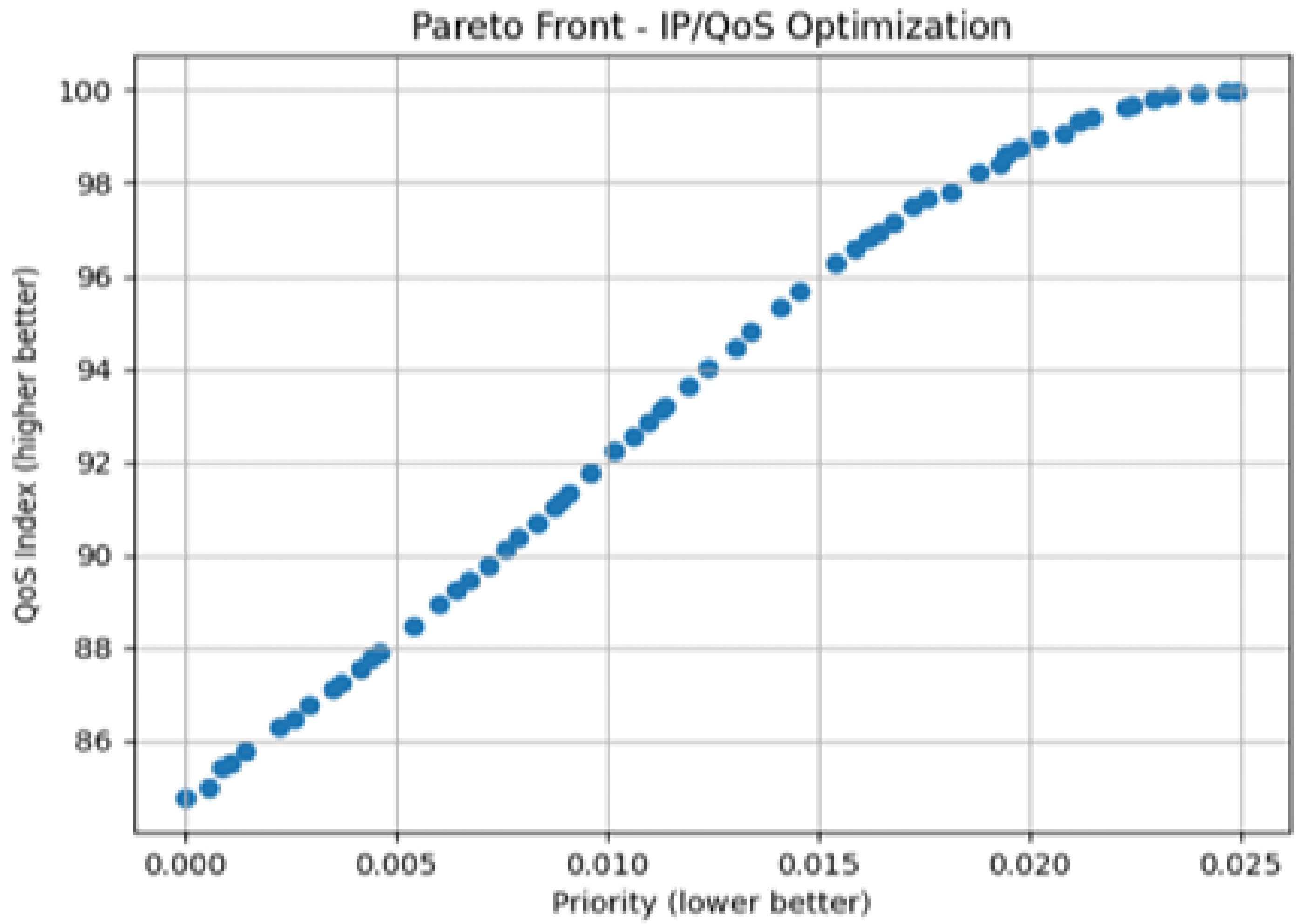


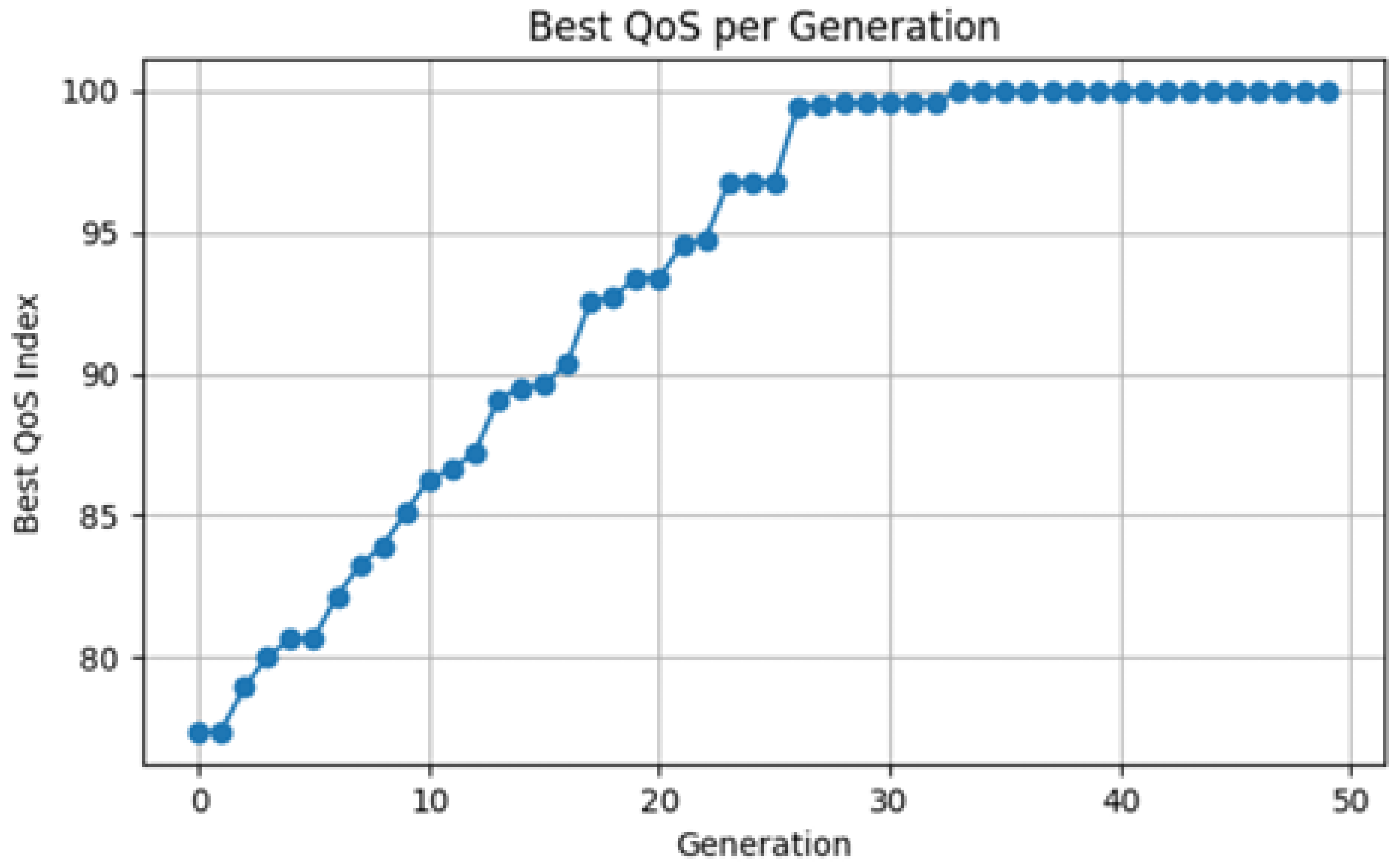
SECURITY THREAT ASSESSMENT

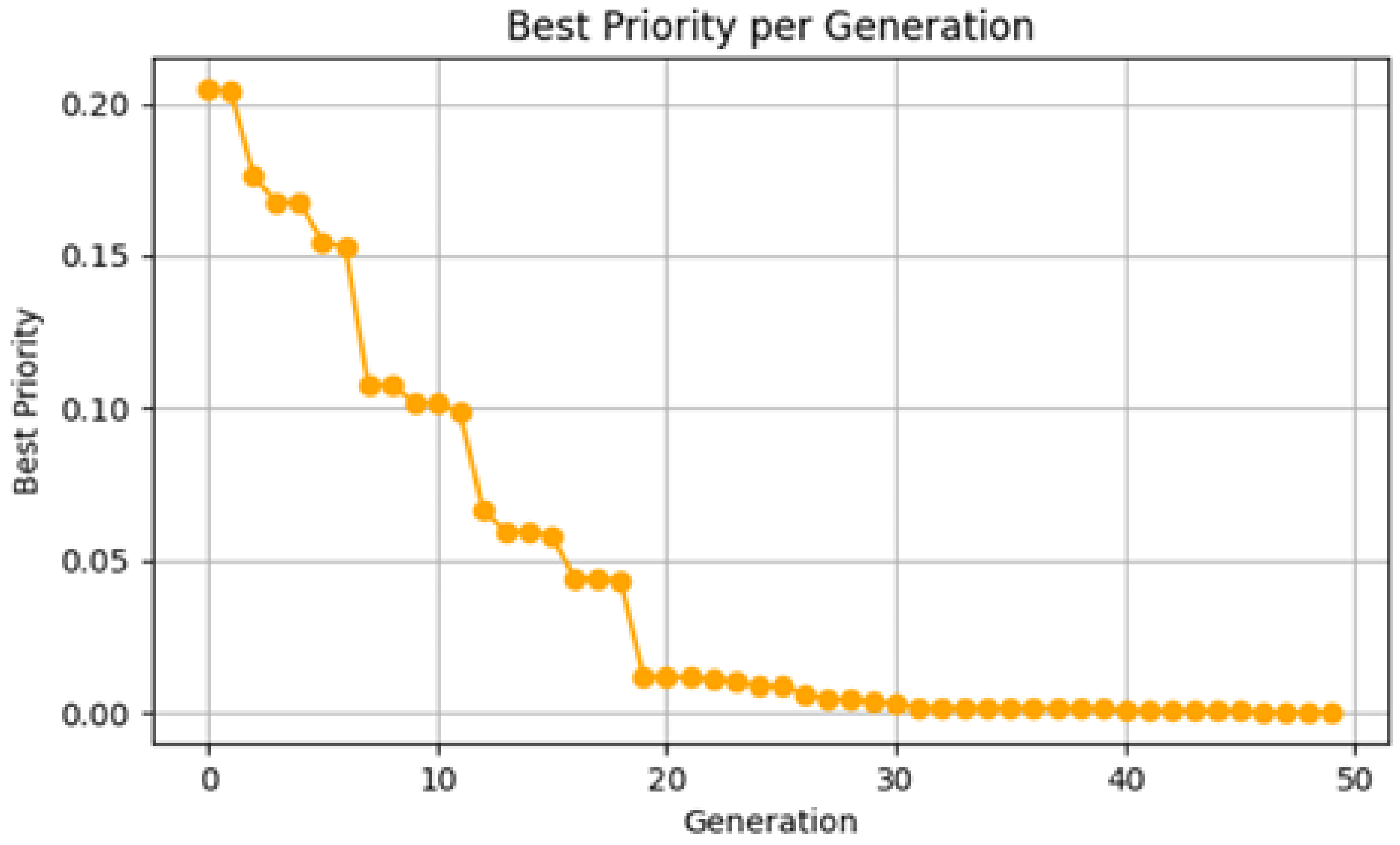




IP AND QOS MANAGEMENT







CONCLUSION & FUTURE WORK

Conclusion

✔️ Hybrid AI framework for intelligent network management

✔️ Improved QoS, VPN selection, and threat mitigation

✔️ Reduced latency and enhanced traffic prioritisation

✔️ ANN + Fuzzy + GA enabled adaptive optimisation

✔️ Scalable solution for real-time network automation

Future Work

✔️Real-time enterprise deployment

✔️ SDN and cloud integration

✔️ Large-scale IoT implementation

✔️ Deep learning-based threat intelligence

✔️ Self-healing autonomous networks

REFERENCES

[1] R. Mahalakshmi and L. Nithyanandham, “Intelligent network performance optimization using AI-driven modelling,” Simulation Modelling Practice and Theory, vol. 123, 2025.

[2]] S. Long et al., "A survey on intelligent network operations and performance optimization based on large language models," IEEE Commun. Surveys Tuts., early access, 2025, doi: 10.1109/COMST.2025.3526606.

[3] S. Ge and Y. Yin, "Research on computer network optimization and reliability enhancement based on improved genetic algorithm," in Proc. IEEE 4th ICDSCA, 2024, pp. 590–594.

[4] B. Basumatary, C. Kumar, and D. K. Yadav, “Security Risk Assessment of Information Systems in an Indeterminate Environment,” Proc. 11th Int. Conf. CONFLUENCE 2021, Amity University, Greater Noida, India, Jan. 2021.

[5] A. Ali et al., "Enhancing cybersecurity with artificial neural networks: A study on threat detection and mitigation strategies," in Proc. 2nd ICCR, 2024, pp. 1–5.

[6] Z. Yang et al., "A malware detection method based on genetic algorithm optimized CNN–SENet network," IEEE Access, vol. 12, pp. 160052–160063, 2024.

[7] S. B. Bose and S. S. Sujatha, "Fuzzy logic-based QoS management and monitoring system for cloud computing," in Proc. 3rd ICISS, 2020, pp. 920–924.

[8] T. M. Amirthalakshmi, S. Raja, M. Subramanian, C. Nelson, S. Rajkumar, R. Makala, and W. Dinku, “QoS Management in Industrial Mobile Communication Using PSO-Based Vision Technique,” Wireless Communications and Mobile Computing, 2023

[9] K. A. Tabary et al., "QoS aware task scheduling using hybrid genetic algorithm in cloud computing," IEEE Access, vol. 13, pp. 51603–51616, 2025.

[10] D. Abhiram, R. Harish, and K. Praveen, Zero-Trust Security Implementation Using SDP over VPN, Inventive Communication and Computational Technologies, Lecture Notes in Networks and Systems, Springer, 2022.

[11] I. Kotenko and I. Saenko, "The genetic approach for design of virtual private networks," in Proc. IEEE 18th CSE, 2015, pp. 168–175.

[12] S. R. Hegde, V. V., K. R. M. V. Chandrakala, G. K. T. and V. K. A. Shankar, "Enhancing Anomaly Detection in Electric Vehicle Supply Equipment (EVSE) Networks Using Classical and Ensemble Learning Approaches," 2024 Control Instrumentation System Conference (CISCON), Manipal, India, 2024, pp. 1-5, doi: 10.1109/CISCON62171.2024.10696830.

[13] S. R. J. Varghese, A. Venu, K. R. M. Vijaya Chandrakala, T. Gireesh Kumar and V. K. Arun Shankar, "Enhancing Power Grid Security by Detecting Relay Setting Manipulation Attacks with Machine Learning," 2024 3rd International Conference on Artificial Intelligence For Internet of Things (AIIoT), Vellore, India, 2024, pp. 1-6, doi: 10.1109/AIIoT58432.2024.10574573.

[14] M. Nakkeeran, V. Anantha Narayanan, P. Bagavathi Sivakumar, and S. Balamurugan, “Decentralized cybersecurity in smart grids: Leveraging location-FedAvg for rapid threat detection and adaptive resilience,” Results in Engineering, 2026, Art. no. 108518, doi: 10.1016/j.rineng.2025.108518.

[15] V. R. V. Lakshmi and T. Gireesh Kumar, “Enhancing data integrity in opportunistic mobile social network: Leveraging Berkle tree and secure data routing against attacks,” Computers & Security, 2025, Art. no. 104133, doi: 10.1016/j.cose.2024.104133

THANK YOU

Questions & Discussions

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